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Roll No.

220824

**2nd Sem / Computer , Computer
(For Speech and Hearing Impaired)
Subject : Analog Electronics**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Zener diode is used as . (CO1)

- a) an amplifier b) voltage regulator
- c) coupler d) rectifier

Q.2 FET is essentially a (CO2)

- a) Current driven device
- b) Voltage driven device
- c) Power driven device
- d) None

Q.3 Minority carriers in N-type semiconductors is. (CO1)

- a) Electrons b) Holes
- c) Both d) None

Q.4 An oscillator requires (CO4)

- a) Positive feedback
- b) Negative feedback
- c) An amplifier and positive feedback
- d) An amplifier and negative feedback

Q.5 Applications of diode (CO5)

- a) Clampers b) Clippers
- c) Rectifiers d) All of the above

Q.6 How many diodes does a single-phase bridge rectifier consists of? (CO3)

- a) One b) Two
- c) Three d) Four

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define forbidden energy gap? (CO1)

Q.8 Draw symbol of NPN transistor. (CO2)

Q.9 What is rectifier? (CO3)

Q.10 JFET stands for (CO2)

Q.11 Define CMRR (CO5)

Q.12 OP-AMP _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Difference between P and N type semiconductor. (CO1)

Q.14 Compare JFET and BJT. (Co2)

Q.15 What is the significance of ripple factor. (CO3)

Q.16 What is feedback? Write advantages of feedback. (CO4)

Q.17 Explain the working of 555 timer as astable multivibrator. (CO5)

Q.18 What is need of filter circuit. Explain LC filter circuit. (CO3)

Q.19 Explain the working of Class 'A' amplifier. (CO4)

Q.20 Explain working of transistor as an amplifier. (CO2)

Q.21 Explain working of Zener diode. (CO1)

Q.22 Draw half wave rectifier and explain its working. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Write short note on : (CO4)

- a) Push Pull amplifier
- b) RC phase shift oscillator

Q.24 Explain V-I characteristics of p-n junction diode. (CO1)

Q.25 Draw and explain input characteristics of transistor in CB configuration. (CO2)